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CHEMISTRY
9202/2

Paper 2

Mark scheme

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Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Examiner.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

Further copies of this mark scheme are available from oxfordaqa.com

Level of response marking instructions

Level of response mark schemes are broken down into levels, each of which has a descriptor. The descriptor for the level shows the average performance for the level. There are marks in each level.

Before you apply the mark scheme to a student's answer read through the answer and annotate it (as instructed) to show the qualities that are being looked for. You can then apply the mark scheme.

Step 1 Determine a level

Start at the lowest level of the mark scheme and use it as a ladder to see whether the answer meets the descriptor for that level. The descriptor for the level indicates the different qualities that might be seen in the student's answer for that level. If it meets the lowest level then go to the next one and decide if it meets this level, and so on, until you have a match between the level descriptor and the answer. With practice and familiarity you will find that for better answers you will be able to quickly skip through the lower levels of the mark scheme.

When assigning a level you should look at the overall quality of the answer and not look to pick holes in small and specific parts of the answer where the student has not performed quite as well as the rest. If the answer covers different aspects of different levels of the mark scheme you should use a best fit approach for defining the level and then use the variability of the response to help decide the mark within the level, ie if the response is predominantly level 3 with a small amount of level 4 material it would be placed in level 3 but be awarded a mark near the top of the level because of the level 4 content.

Step 2 Determine a mark

Once you have assigned a level you need to decide on the mark. The descriptors on how to allocate marks can help with this. The exemplar materials used during standardisation will help. There will be an answer in the standardising materials which will correspond with each level of the mark scheme. This answer will have been awarded a mark by the Lead Examiner. You can compare the student's answer with the example to determine if it is the same standard, better or worse than the example. You can then use this to allocate a mark for the answer based on the Lead Examiner's mark on the example.

You may well need to read back through the answer as you apply the mark scheme to clarify points and assure yourself that the level and the mark are appropriate.

Indicative content in the mark scheme is provided as a guide for examiners. It is not intended to be exhaustive and you must credit other valid points. Students do not have to cover all of the points mentioned in the Indicative content to reach the highest level of the mark scheme.

An answer which contains nothing of relevance to the question must be awarded no marks.

Information to Examiners

1. General

The mark scheme for each question shows:
- the marks available for each part of the question
- the total marks available for the question
- the typical answer or answers which are expected
- extra information to help the Examiner make his or her judgement
- the Assessment Objectives and specification content that each question is intended to cover.

The extra information is aligned to the appropriate answer in the left-hand part of the mark scheme and should only be applied to that item in the mark scheme.

At the beginning of a part of a question a reminder may be given, for example: where consequential marking needs to be considered in a calculation; or the answer may be on the diagram or at a different place on the script.

In general the right-hand side of the mark scheme is there to provide those extra details which confuse the main part of the mark scheme yet may be helpful in ensuring that marking is straightforward and consistent.

2. Emboldening and underlining

2.1 In a list of acceptable answers where more than one mark is available 'any **two** from' is used, with the number of marks emboldened. Each of the following bullet points is a potential mark.

2.2 A bold **and** is used to indicate that both parts of the answer are required to award the mark.

2.3 Alternative answers acceptable for a mark are indicated by the use of **or**. Different terms in the mark scheme are shown by a / ; eg allow smooth / free movement.

2.4 Any wording that is underlined is essential for the marking point to be awarded.

3. Marking points

3.1 Marking of lists

This applies to questions requiring a set number of responses, but for which students have provided extra responses. The general principle to be followed in such a situation is that 'right + wrong = wrong'.

Each error / contradiction negates each correct response. So, if the number of errors / contradictions equals or exceeds the number of marks available for the question, no marks can be awarded.

However, responses considered to be neutral (indicated as * in example 1) are not penalised.

Example 1: What is the pH of an acidic solution?

[1 mark]

Student	Response	Marks awarded
1	green, 5	0
2	red*, 5	1
3	red*, 8	0

Example 2: Name two planets in the solar system.

[2 marks]

Student	Response	Marks awarded
1	Neptune, Mars, Moon	1
2	Neptune, Sun, Mars, Moon	0

3.2 Use of chemical symbols / formulae

If a student writes a chemical symbol / formula instead of a required chemical name, full credit can be given if the symbol / formula is correct and if, in the context of the question, such action is appropriate.

3.3 Marking procedure for calculations

Marks should be awarded for each stage of the calculation completed correctly, as students are instructed to show their working. Full marks can, however, be given for a correct numerical answer, without any working shown.

3.4 Interpretation of ‘it’

Answers using the word ‘it’ should be given credit only if it is clear that the ‘it’ refers to the correct subject.

3.5 Errors carried forward

Any error in the answers to a structured question should be penalised once only.

Papers should be constructed in such a way that the number of times errors can be carried forward is kept to a minimum. Allowances for errors carried forward are most likely to be restricted to calculation questions and should be shown by the abbreviation ecf in the marking scheme.

3.6 Phonetic spelling

The phonetic spelling of correct scientific terminology should be credited **unless** there is a possible confusion with another technical term.

3.7 Brackets

(…) are used to indicate information which is not essential for the mark to be awarded but is included to help the examiner identify the sense of the answer required.

3.8 Allow

In the mark scheme additional information, ‘allow’ is used to indicate creditworthy alternative answers.

3.9 Ignore

‘Ignore’ is used when the information given is irrelevant to the question or not enough to gain the marking point. Any further correct amplification could gain the marking point.

3.10 Do <u>not</u> accept

‘Do **not** accept’ means that this is a wrong answer which, even if the correct answer is given as well, will still mean that the mark is not awarded.

Question 1

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.1	**B**		1	AO1 3.7.2a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.2	**A**		1	AO1 3.1.3a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.3	**C**		1	AO1 3.1.3a 3.7.1c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.4	**D**		1	AO1 3.2.1d 3.7.1a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.5	form coloured compounds		1	AO1 3.7.1a 3.7.2a,b

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.6	63 (°C)	allow a value in the range 53–73 °C	1	AO3 3.1.3a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.7	(going down Group 1) the melting points decrease		1	AO3 3.1.3a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.8	(metal) **Z** (reason) has ions with different charges acts as a catalyst	if no other mark awarded, allow for **1** mark metal **X** acts as a catalyst	1 1 1	AO3 3.7.2a

Total Question 1		**10**

Question 2

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.1	**A**		1	AO2 3.10.1.2c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.2	ethane		1	AO1 3.10.1.2b

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.3	covalent		1	AO1 3.2.1g 3.10.1.2b

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.4	C_nH_{2n+2}		1	AO1 3.10.1.2a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.5	evaporate condense		1 1	AO1 3.10.1.1c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.6	petrol		1	AO2 3.10.1.1c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.7	heavy fuel oil		1	AO2 3.10.1.2c

Total Question 2		**8**

Question 3

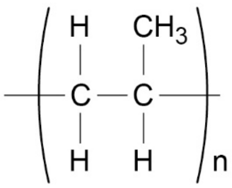
Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.1	flammable substances	allow 'suck back'	1	AO4 3.10.1.2c 3.10.1.3a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.2	decomposition		1	AO1 3.10.1.3a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.3	C_2H_4		1	AO2 3.10.1.3b 3.10.1.3c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.4	(test) bromine (water) (result) (bromine water turns from) orange to colourless	 allow (bromine water) decolourises MP2 dependent on MP1 being awarded	 1 1	AO1 3.10.1.3d RPA 5

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.5	fuel		1	AO1 3.10.1.3e

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.6	C–C bond 3 × C–H bonds **and** 1 × C–CH_3 open end bonds **and** n	an answer of  scores **3** marks	1 1 1	AO2 3.10.2a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.7	(thermosetting polymers) consist of polymer chains with cross links between them which take a lot of energy to break		1 1	AO2 3.10.2c

Question	Answers	Mark	AO/ Spec. Ref.
03.8	**Level 3**: A judgement, strongly linked and logically supported by a sufficient range of correct reasons, is given.	5–6	AO3 3.10.1.2g 3.10.2d 3.10.2e
	Level 2: Some logically linked reasons are given. There may also be a simple judgement.	3–4	
	Level 1: Relevant points are made. They are not logically linked.	1–2	
	No relevant content	0	
	Indicative content: • wood is obtained from trees which are a renewable resource • poly(chloroethene) is made from crude oil which is a non-renewable resource • wood releases less carbon dioxide during production • so lower carbon footprint • using wood has impact on land use • so less land for food production • wood needs to be treated for weather resistance • which adds to cost • wood is less durable • so needs replacing more often **or** treating regularly • wood biodegrades • so easier to dispose of • a reasoned judgement		

Total Question 3		**17**

Question 4

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.1	sodium (atom) loses electron(s)		1	AO1 3.2.1b,c 3.7.1a,c
	chlorine (atom) gains electron(s)		1	
	reference to one electron		1	
	sodium ions **and** chloride ions are formed	allow Na^+ ions **and** Cl^- ions are formed allow sodium forms positive ions **and** chlorine forms negative ions allow to form ions with a complete outer shell of electrons	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.2	(percentage =) $\frac{23}{58.5} \times 100$		1	AO2 3.6.2b
	= 39.316		1	
	= 39	allow an answer correctly rounded to 2 significant figures from a calculation which uses the values in the question	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.3	(sodium chloride) has a giant (ionic) structure **or** (sodium chloride) has a regular structure	allow (sodium chloride) has a lattice (structure)	1	AO2 3.2.2a
	(which has strong) electrostatic forces of attraction between oppositely charged ions		1	
	(which) act in all directions		1	
	(that) take large amounts of energy to break		1	

Total Question 4		**11**

Question 5

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.1	production of copper is increasing		1	AO3 3.3.1c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.2	increase in population / demand	allow more uses for copper	1	AO3 3.3.1c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.3	bacteria		1	AO1 3.3.1.1f

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.4	iron is more reactive than copper		1	AO2 3.3.1.1b,h

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.5	iron loses electrons (to form Fe^{2+} ions) copper ions gain electrons (to form copper atoms) this is a redox reaction because oxidation and reduction are occurring (at the same time) (as) losing electrons is oxidation **and** gaining electrons is reduction		1 1 1 1	AO1 × 1 AO2 × 3 3.3.1.1h 3.8.4b 3.8.4c

Total Question 5		**8**

Question 6

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.1	(improvement) use a pipette (instead of a measuring cylinder) (reason) more accurate (volume measured)	allow use a burette (instead of a measuring cylinder)	1 1	AO4 3.6.4b RPA 3

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.2	(universal indicator) gives a range of colours (so) is hard to judge an end point		1 1	AO4 3.5.1f 3.6.4b RPA 3

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.3	use of concordant results 18.05, 17.95 and 18.05 only (mean volume of sulfuric acid =) $\frac{18.05 + 17.95 + 18.05}{3}$ = 18.02 (cm^3)	 allow correct use of incorrectly determined concordant results allow 18.01666 (cm^3)	1 1 1	AO2 × 2 AO3 × 1 3.6.4c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.4	(volume conversion of sulfuric acid = $\frac{19.05}{1000}$ =) 0.01905 (dm^3)		1	AO2 3.6.4c
	(moles of sulfuric acid = 0.01905 × 0.30 =) 5.715×10^{-3}	allow correct use of an incorrect / no conversion of sulfuric acid volume	1	
	(moles of sodium hydroxide = $5.715 \times 10^{-3} \times 2$ =) 1.143×10^{-2}	allow correct use of incorrectly determined moles of sulfuric acid	1	
	(concentration = $\frac{1.143 \times 10^{-2}}{0.0250}$ =) 0.4572 (mol/dm^3)	allow correct use of incorrectly determined moles of sodium hydroxide	1	

Total Question 6		**11**

Question 7

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.1	(solution **X**) limewater (gas) carbon dioxide	allow calcium hydroxide solution allow CO_2	1 1	AO2 3.3.1.2a 3.4.2c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.2	(green) copper carbonate (thermally) decomposes to produce black copper oxide		1 1	AO3 3.3.1.2a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.3	(metal carbonate) potassium carbonate (reason) the temperatures reached by the Bunsen burner are insufficient	allow K_2CO_3 allow sodium carbonate	1 1	AO1 × 1 AO2 × 1 3.3.1.2a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.4	(*x*-axis scale) 0.20, 0.40, 0.60, 0.80, 1.00 all points correctly plotted line of best fit	ignore intermediate values allow a tolerance of ± ½ a small square allow **1** mark for three or four points correctly plotted	1 2 1	AO2 × 3 AO3 × 1 3.3.1.2b

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.5	a delay in connecting the stopper to the conical flask (which) caused some gas to escape		1 1	AO3 3.3.1.2b

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.6	volume of gas collected		1	AO4 3.3.1.2b

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.7	the same volume of gas collected (because) there are the same number of moles of calcium carbonate	allow volume of gas collected was 125 cm^3	1 1	AO3 3.6.1c

Total Question 7		**15**

Question 8

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.1	zero (error)		1	AO4 3.9.2a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.2	(temperature change = 52.3 – 18.7 =) 33.6 (°C)		1	AO2 3.9.2a,b
	(energy released =) 100 × 4.2 × 33.6	allow correct use of an incorrectly determined temperature change	1	
	= 14 112 (J)		1	
	= 14.112 (kJ)	allow correct conversion of an incorrectly determined value for energy released	1	
	(mass of ethanol burnt = 28.5 – 26.9 =) 1.6 (g)		1	
	(energy released = $\frac{14.112}{1.6}$) = 8.82 (kJ/g)	allow correct use of an incorrectly determined value for energy released in kJ allow correct use of an incorrectly determined mass of ethanol burnt	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.3	reproducible		1	AO4 3.9.2a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.4	energy was transferred to the air (so) less energy was transferred to the water		1 1	AO4 3.9.2a

Total Question 8		**10**